Consciousness: Here, There, and Everywhere?

Based on the paper by Giulio Tononi & Christof Koch (2015) *Phil. Trans. R. Soc. B 370: 20140167*

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A Presentation for Artificial Intelligence and Machine Consciousness Course

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I know I am conscious: I am seeing, hearing, feeling something here, inside my own head. But is consciousness—subjective experience—also there, not only in other people's heads, but also in the head of animals? And perhaps everywhere, pervading the cosmos, as in old panpsychist traditions and in the Beatles' song?

The Science of Consciousness: Current Landscape

- **Significant Progress:** Identifying behavioral and neuronal correlates (NCCs) of experience.
- **But Correlates Aren't Enough:**
  - Don't explain *why* (e.g., Cerebral cortex vs. Cerebellum – more neurons, complex, yet no direct consciousness link).
  - Don't explain *how* physical processes become subjective experience (The "Hard Problem").
- **The Question is Urgent & Expanding:**
  - Clinical: Brain-injured patients, "islands of cortex."
  - Developmental: Infants, fetuses.
  - Non-Human: Animals far removed from us.
  - **Artificial Intelligence:** Increasingly capable machines.

- **Critical Need:** A *theory* of consciousness – what it is & what physical systems can have it.

Assessing Consciousness: The Challenges

1. Behavioral Correlates & Reportability

- **Traditional:** Observation, verbal reports ("gold standard" in labs).
- **Deep Limitations:**
  - Behavior can be misleading (e.g., sleepwalking vs. vivid dreaming).
  - Reportability issues: Non-verbal subjects (infants, animals), unreported experiences (dreams).

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2. Neural Correlates of Consciousness (NCCs)

- **Definition:** Minimal neural mechanisms for a specific conscious percept.
- **The Search is On, But...**
  - **No Consensus:** Many candidates (fronto-parietal activity, gamma waves, P300), but no universal "signature."
  - **Confounding Factors:** Are we seeing correlates of experience, or of processes *around* it (attention, memory, report)?

- **Generalizability:** NCCs from healthy adults may not apply to:
  - Brain-damaged patients.
  - Infants.
  - Different animal species.
  - Machines.

The Really Hard Cases: Where Correlates Fail Us

(This is where the problem becomes acute)

Patients & Infants

- **"Islands of Cortex":** Is a small patch of functioning brain tissue conscious? Of what? (Pain? Sound?)
- **Newborns/Preterm Babies:** Immature brains, limited connectivity. What is their experience, if any? (Absence of adult NCCs isn't definitive).

Animals

- **Beyond Mammals:** Strong arguments for consciousness in mammals (behavioral, neural similarity).
- **What about...** Ravens, octopuses, bees?
  - Sophisticated, learned behaviors.
  - Vastly complex nervous systems (e.g., bee brain: 800k neurons, dense circuits).
- **The "Cerebellum Problem" Analogy:**
  - Human cerebellum: 69 billion neurons (most of the brain's total!), vital for motor control, learning.
  - Loss of cerebellum doesn't mean loss of consciousness.
  - **Is a bee's brain like our cortex (conscious) or cerebellum (complex but not generating experience)?**

Machines & AI

- **Rapid Advancements:** AI outperforms humans in chess (Deep Blue), Q&A (Watson), driving (Google cars), image recognition.
- **The Big Question:** Humans doing these tasks are assumed conscious. What about these digital entities?
  - Radically different architecture (designed, not evolved).
  - **Does computational prowess equal subjective experience?**

Everywhere?

In the ‘Canticle of the Creatures’, Saint Francis addressed animals, flowers and even stones as if endowed with soul, and praised them as mother earth, brother sun, sister moon, the stars, the air, water and fire. And he was not alone.

Philosophical Perspectives: Panpsychism & IIT's Angle

(Briefly touching on broader ideas)

- **Panpsychism (Ancient Idea):** Consciousness/mind is a fundamental, ubiquitous feature of reality.
- **Integrated Information Theory (IIT) offers a lens:**
  - Views consciousness as intrinsic, fundamental (like mass).
  - Suggests consciousness is *graded* (can vary in richness).
  - **BUT: Not everything is conscious.** (Key distinction from some panpsychist views).
    - *Aggregates* (a nation, a heap of sand) are not conscious super-entities.
    - Addresses the "combination problem."

IIT's Angle: Complicated Systems Can Be Unconscious

(Crucial implications for AI)

- **Feed-Forward Networks:**
  - Information flows one way (input -> output).
  - Can be incredibly complex and perform sophisticated tasks.
  - IIT predicts: **Zero consciousness.** (Effectively "zombies").

- **Simulations of Conscious Brains:**
  - Even a perfect digital simulation of a human brain on a conventional computer.
  - IIT suggests: **Likely NOT conscious** like the original.
    - Consciousness requires *real* intrinsic cause-effect power in the *physical system itself*.
    - A simulation is virtual. The computer's hardware (transistors) would likely be the basis for any consciousness, and that would be vastly different and simpler.
- **Key Implication:** Internal architecture matters, not just input-output behavior. **No ultimate Turing Test for consciousness.**

Conclusion & The Path Forward

- **Beyond Correlates:** Understanding consciousness demands a theoretical framework.
- **A Scientific Approach:**
  1. Test theories in accessible situations (e.g., human consciousness changes in sleep, anesthesia).
  2. Validate and refine the theory.
  3. Extrapolate with caution to more remote cases (patients, infants, animals, machines).
- **The Goal:** To make the best inferences possible based on a robust theory, rather than resorting to arbitrary claims or remaining silent on these profound questions.
- **Consciousness remains one of the greatest scientific and philosophical challenges.**

Q&A / Discussion